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the integrated Man of tomorrow

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*** START OF THE PROJECT GUTENBERG EBOOK BRAIN TEASER

BRAIN TEASER

BY TOM GODWIN

*How can a ship travel both forward and
backward and sideways in two different directions,
be going twice as fast as the speed of
light—and still be completely motionless?*

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Carl Engle stood aside as the flight preparation crew filed out of the *Argosy's* airlock. Barnes was the last; fat and bald and squinting against the brightness of the Arizona sun.

"All set, Carl," he said. "They had us to check and countercheck, especially the drives."

Engle nodded. "Good. Ground Control reports the Slug cruiser still circling seven hundred miles out and they think the Slugs suspect something."

"Damned centipedes!" Barnes said. "I still say they're telepathic." He looked at his watch. Zero hour minus twenty-six minutes. "Good luck, boy, and I hope this space warp dingus works like they think it will."

He waddled down the boarding ramp and Engle went through the airlock, frowning a little as he threw the switches that would withdraw the ramp and close the airlock behind him. Barnes' implied doubt in the success of the space warp shuttle was not comforting. If the shuttle failed to work, the *Argosy* would be on the proverbial spot with the Slug cruiser eager to smear it well thereupon....

Access to the control room was up through the room that housed the space warp shuttle. Dr. Harding, the tall, bristle-browed physicist, and his young assistant, Garvin, looked up briefly as he entered then returned their attention to their work. The master computer, borrowed from M.I.T., stood like a colossal many-dialed refrigerator along one wall. A protective railing around it bore a blunt KEEP OUT sign and it was never left unwatched. Garvin was seated before it, his fingers flitting over the keyboard and the computer's answer panel replying with strange mathematical symbols.

The space warp shuttle sat in the middle of the room, a cube approximately two-thirds of a meter along the edge, studded with dials and knobs and surmounted by a ball of some shining silvery alloy. Dr. Harding was talking into the transdimensional communicator mounted beside the shuttle.

Engle went on to the computer and waited outside the railing until Garvin finished with his work and turned in his seat to face him.

"The last check question," Garvin said. "Now to sweat out the last twenty minutes."

"If you've got the time, how about telling me about the shuttle," said Engle, "I've been kept in the dark about it; but from what I understand, the shuttle builds up a field around the ship, with the silver ball as the center of the field, and this field goes into another dimension called the 'space warp'."

"Ah—it could be described in that manner," Garvin said, smiling a little. "A clear description could not be made without the use of several special kinds of mathematics, but you might say this field in normal space is like a bubble under water. The air bubble seeks its own element, rises rapidly until it emerges into free air—in this case, the space warp. This transition into the warp is almost instantaneous and the shuttle automatically ceases operation when the warp is fully entered. The shuttle is no longer needed; the hypothetical bubble no longer exists—it has found its own element and merged with it."

"I know that a light-hour of travel in the warp is supposed to be equivalent to several light-years in normal space," Engle said, "but what about when you want to get back into normal space?"

"The original process is simply reversed: the shuttle creates a 'bubble' that cannot exist in the warp and seeks its own element, normal space."

"I see. But if the shuttle should—"

He never completed the question. Dr. Harding strode over, his eyes blue and piercing under the fierce eyebrows as he fixed them on him. He spoke without preamble:

"You realize the importance of this test flight with the shuttle, of course? Entirely aside from our personal survival should the Slug cruiser intercept us."

"Yes, sir," he answered, feeling the question suggested an even lower opinion of his intelligence than he had thought Harding held.

Project Space Warp existed for the purpose of sending the *Argosy* to Sirius by means of the space warp shuttle and bringing back the *Thunderbolt* by the same swift method. The *Thunderbolt*, Earth's first near-to-light-speed interstellar ship, was a huge ship; armed, armored, and invincible. It had been built to meet every conceivable danger that might be encountered in interstellar exploration—but the danger had come to the solar system from the direction of Capella nine years after the departure of the *Thunderbolt*.

Eight cruisers of the pulpy, ten-foot centipede-like things called Slugs had methodically destroyed the colonies on Mars and Venus and established their own outposts there. Earth's ground defenses had held the enemy at bay beyond the atmosphere for a year but such defense could not be maintained indefinitely. The *Thunderbolt* was needed quickly and its own drives could not bring it back in less than ten years....

"We will go into the warp well beyond the atmosphere," Harding said. "Transition cannot be made within an atmosphere. Since a very moderate normal space velocity of the ship will be transformed into a greater-than-light velocity when in the warp, it is desirable that we make turn-over and decelerate to a very low speed before going into the warp."

"Yes, sir," he said. "I was briefed on that part and I'll bring us as near to a halt as that cruiser will permit."

"There will be communication between us during the flight," Harding said. "I will give you further instructions when they become necessary."

He turned away with an air of dismissal. Engle went to the ladder by the wall. He climbed up it and through the interroom airlock, closing the airlock behind him; the routine safety measure in case any single room was punctured. He went to the control board with a vague resentment gnawing for the first time at his normally placid good nature.

So far as Harding was concerned—and Garvin, too—he might as well have been an unusually intelligent baboon.

Zero hour came and the *Argosy* lifted until Earth was a tremendous, curving ball below and the stars were brilliant points of light in a black sky. The Slug cruiser swung to intercept him within the first minute of flight but it seemed to move with unnatural slowness. It should have been driving in at full speed and it wasn't....

"Something's up," Ground Control said. "It's coming in too slowly."

"I see that," he answered. "It must be covering something beyond it, in your radar shadow."

It was. When he was almost free of the last traces of atmosphere he saw the other cruiser, far out and hidden from Ground Control's radar by the radar shadow cast by the first one.

He reported, giving its position and course as given him by the robot astrogating unit.

"We'll have the greatest amount of time if I make turn-over now and decelerate," he finished.

The voice of Harding came through the auxiliary speaker:

"Do so."

The *Argosy* swung, end for end, and he decelerated. The cruiser behind him increased its speed, making certain it would be in position to cut off any return to Earth. The other cruiser altered its course to intersect the point in space the *Argosy* would soon occupy, and the *Argosy* was between the rapidly closing jaws of a trap.

He made reports to Ground Control at one-minute intervals. At 11:49 he said:

"Our velocity is approaching zero. We'll be within range of the second cruiser's blasters in two more minutes."

Harding spoke again to him:

"We'll go into the warp now. *Do not* alter the deceleration or the course of the ship while we're in the warp."

"I won't," he said.

There was a faint mutter from the auxiliary speaker as Harding gave some instructions to Garvin. Engle took a last look at the viewscreen; at blue-green Earth looming large in the center, Orion and Sirius glittering above it and the sun burning bright and yellow on the right. It was a scene he had observed many times before, all very familiar and normal—

The chronometer touched 11:50 and normalcy vanished.

Earth and sun and stars fled away from him, altering in appearance as they went, shrinking, dwindling. The seas and continents of Earth erupted and shook and boiled before Earth faded and disappeared. The sun changed

from yellow to green to blue, to a tiny point of bright violet light that raced away into the blackness filling the screen and faded and disappeared as Earth had done.

Then the viewscreen was black, utterly, completely, dead black. And the communicator that had connected him with Ground Control was silent, without the faintest whisper of background sound or space static.

In the silence the voice of Harding as he spoke to Garvin came through the speaker; puzzled, incredulous, almost shocked:

"Our velocity couldn't have been that great—and *the sun receded into the ultraviolet!*"

There was the quick sound of hurrying footsteps then the more distant sound of the computer's keys being operated at a high rate of speed. He wanted to ask what had gone wrong but he knew no one would answer him. And it would be a pointless question—it was obvious from Harding's tone that he did not know, either.

He had an unpleasant feeling that Man's first venture into another dimension had produced catastrophic results. What had caused sun and Earth to disappear so quickly—and what force had riven and disfigured Earth?

Then he realized the significance of Harding's statement about the sun receding into the ultraviolet.

If the ship had been traveling at a high velocity away from the sun, the wave length of the sun's light would have been increased in proportion to the speed of the ship. The sun should have disappeared in the long-wave infrared end of the spectrum, not the short-wave ultraviolet.

With the thought came the explanation of the way the continents and oceans of Earth had quivered and seethed. The shifting of the spectrum range had shortened normally visible rays into invisibly short ultraviolet radiations while at the same time formerly invisible long infrared radiations had been shortened into visible wave lengths. There had been a continuous displacement into and past the ultraviolet and each wave length would have reflected best from a different place—mountains, valleys, oceans, deserts, warm areas, cool areas,—and the steady progression into the ultraviolet had

revealed each area in quick succession and given the appearance of agitated movement.

So there was no catastrophe and everything had a logical explanation. Except how they could have been approaching a sun that he had seen clearly, visibly, racing away from them.

"Engle—" The voice of Harding came through the speaker. "We're going back into normal space to make another observation. I don't know just where we are but we're certain to be far from the cruisers. Don't alter our course or velocity."

"Yes, sir," he said.

They came out of the warp at 11:53. The communicator burped suddenly and the viewscreen came to life; a deep, dull red that brightened quickly. A tiny coal flared up, swelling in size and shifting from red to orange to yellow—the sun. Earth appeared as a hazy red dot that enlarged and resolved itself into a planet with distorted continents that trembled and changed, to resume their natural shapes and colors. Within a few seconds the sun was shining as ever, Earth loomed large and blue-green before them and the stars of Orion glittered unchanged beyond. Even their position in space was the same—they had not moved.

But the Slug cruisers had.

One was very near and from its forward port came the violet haze that always preceded a blaster beam. There was no time to escape—no chance at all. He spoke into the mike, harsh and urgent:

"Into the warp! There's a blaster beam coming—move!"

There was a silence from below that seemed to last an eternity, then the sound of a switch being slapped hastily. At the same time, the violet haze before the cruiser erupted into blue fire and the blaster beam lanced out at them.

It struck somewhere astern. The power output needle swung jerkily as the generators went out and the emergency batteries took the heavy load of the shuttle's operation. There was a sensation of falling as the ship's artificial gravity units ceased functioning. The auxiliary speaker rattled wordlessly

and there was a sound like a hard rush of wind through it, accompanied by quick bumping sounds.

Then the speaker was still and there was no sound of any kind as the viewscreen shifted into the ultraviolet and Earth and stars and sun once again raced away and disappeared in the blackness.

A myriad of lights above the board informed him the generators were destroyed, the stern section riddled and airless, the emergency batteries damaged and reduced to quarter charge, the shuttle room punctured and airless.

And, of course, Harding and Garvin were dead.

He felt a surge of futile anger. It had all been unnecessary. If only they had not considered him incompetent to be entrusted with anything more than the ship's operation—if only they had installed an emergency switch for the shuttle by his control board, there would not have been the two-second delay following his order and they would have been safely in the warp before the blaster beam struck.

But they had not trusted him with responsibility and now he was alone in a space warp he did not understand; sole and full responsibility for the shuttle suddenly in his hands.

He considered his course of action, then got into a pressure suit. Magnets in the soles of its heavy boots permitted him to walk in the absence of gravity and he went to the interroom airlock with metallically clicking steps. He let himself through the lock and walked down what had been the room's wall, then across to the center of its floor.

But for the fact there was no one in the room, it was as he had last seen it. The shuttle, computer, and other equipment stood in their orderly positions with their lighted dials unchanged. Until one looked at the gash ripped in the hull and saw the stains along its edge where the occupants had been hurled through it by the escaping air.

He went on to the next room and the next. The damage increased as he proceeded toward the stern. The power generators were sliced into ribbons and the emergency batteries in such condition it seemed a miracle they were functioning at all. The drives had received the greatest damage; they were an unrecognizable mass of wreckage.

He made his way back to the shuttle room, there to appraise his circumstances. He reached automatically for a cigarette and stopped when his glove bumped the breast plate of his pressure suit.

First, he would have to make the shuttle room livable; get out of the pressure suit. He would have to question the computer and he could not do that with the thick, clumsy gloves on his hands.

The job didn't take long. There were repair plates on the ship and a quick-hardening plastic spray. He closed the sternward airlock when he was done and opened the airlock leading to the control room, as well as the locks beyond. Air filled the shuttle room, with only a minor over-all loss of air pressure. He removed the suit, attached a pair of magnetic soles to his shoes so he could operate the keys of the computer without the movements sending him floating away, and went to it.

He had never been permitted to touch it before, nor even stand close enough to see what the keyboard looked like. Now, he saw that the alphabetical portion of the keyboard was minor compared with the mathematical portion, many of the symbols strange to him.

The operation of an interplanetary ship required a certain knowledge of mathematics, but not the kind used by theoretical physicists. He typed, doubtfully:

ARE YOU CAPABLE OF ANSWERING QUESTIONS PRESENTED IN NON-MATHEMATICAL FORM?

The word, YES, appeared at once in the answer panel and relief came to him like the lifting of a heavy burden.

The computer knew as much about the space warp as Harding or anyone else. It was connected with his drive controls and instruments and knew how far, how fast, and in what directions the flight had taken place. It had even been given blueprints of the ship's construction, in case the structure of

the ship should affect the ship's performance in the warp, and knew every nut, bolt, plate and dimension in the ship.

There was supposed to be a certain method of procedure when questioning the computer. "It knows—but it can't think," Garvin had once said. "It lacks the initiative to correlate data and arrive at conclusions unless the procedure of correlation is given it in detail."

Perhaps he could manage to outline some method of correlation for the computer. The facts of his predicament were simple enough:

He was in an unknown medium called "the Space Warp." Something not anticipated occurred when a ship went into the warp and Harding had not yet solved the mystery when he died. The physicists in Observation would be able to find the answer but he could not ask them. The forward movement of the ship was not transferred with it into the warp and if he emerged into normal space the waiting Slug cruisers would disintegrate him before he spoke three words to Observation.

There was a pencil and a tablet of paper by the computer. He used them to calculate the time at which the charge in the damaged batteries would reach a critical low, beyond which the charge would be insufficient to activate the shuttle.

The answer was 13:53. He would have to go out of the warp at 13:53 or remain in it forever. He had a great deal less than two hours in which to act.

He typed the first question to the computer:

WHAT IS THE POSITION OF THIS SHIP RELATIVE TO NORMAL SPACE?

The answer appeared on the panel at once; the coordinates of a position more than a light-year toward Ophiuchus.

He stared at the answer, feeling it must be an error. But it could not be an error—the computer did not make mistakes. How, then, could the ship have traveled more than a light-year during its second stay in the warp when it had not moved at all during the first stay? Had some factor of the warp unknown to him entered the picture?

As a check he typed another question:

WHAT WAS OUR POSITION, RELATIVE TO NORMAL SPACE, IMMEDIATELY BEFORE THIS SHIP WAS SHUTTLED BACK OUT OF THE WARP?

The answer was a position light-days toward Ophiuchus.

He typed: IMPOSSIBLE.

The computer replied: THIS STATEMENT CONFLICTS WITH PREVIOUS DATA.

He recalled the importance of keeping the computer free of all faulty or obscure data and typed quickly: CANCEL CONFLICTING STATEMENT.

CONFLICTING STATEMENT CANCELED, it replied.

He tried another tack. THIS SHIP EMERGED FROM THE SPACE WARP INTO THE SAME NORMAL SPACE POSITION IT HAD OCCUPIED BEFORE GOING INTO THE WARP.

He thought the computer would proceed to give him some sort of an explanation. Instead, it non-committally replied: DATA ACKNOWLEDGED.

He typed: EXPLAIN THIS DISCREPANCY BETWEEN SPACE WARP AND NORMAL SPACE POSITIONS.

It answered: INSUFFICIENT DATA TO ACCOUNT FOR DISCREPANCY.

He asked, HOW DID YOU DETERMINE OUR PRESENT POSITION?

It replied: BY TRIANGULATION, BASED ON THE RECESSION OF EARTH, THE SUN, SIRIUS, ORION, AND OTHER STARS.

BUT THE RECEDING SUN WENT INTO THE ULTRAVIOLET, he objected.

Again it answered with the non-committal, DATA ACKNOWLEDGED.

DID YOU ALREADY HAVE THIS DATA? he asked.

YES.

EXPLAIN WHY THE RECEDING SUN SHIFTED INTO THE ULTRAVIOLET INSTEAD OF THE INFRARED.

It replied: DATA INSUFFICIENT TO ARRIVE AT LOGICAL EXPLANATION.

He paused, pondering his next move. Time was speeding by and he was learning nothing of value. He would have to move the ship to some place in the warp where emergence into normal space would not put him under the blasters of the Slug cruisers. He could not know where to move the ship until he knew where the ship was at the present. He did not believe it was in the position given him by the computer, and its original space warp position had certainly not been the one given by the computer.

The computer did not have the ability to use its knowledge to explain contradictory data. It had been ordered to compute their space warp position by triangulation of the receding sun and stars and was not at all disturbed by the contradicting shift of the sun into the ultraviolet. Suppose it had been ordered to calculate their position by computations based on the shift of the sun's and stars' spectrum into the ultraviolet?

He asked it: WHAT IS OUR POSITION, IGNORING THE TRIANGULATION AND BASING YOUR COMPUTATIONS ON THE SHIFT OF THE SPECTRUMS OF THE SUN AND ORION INTO THE ULTRAVIOLET?

It gave him the coordinates of a position almost two light-years toward Orion. The triangulation computations had shown the ship to be going backward at many times the speed of light; the spectrum-shift computations showed it to be going forward with approximately the same speed.

THIS SHIP CANNOT SIMULTANEOUSLY BE IN TWO POSITIONS THREE LIGHT-YEARS APART. NEITHER CAN IT SIMULTANEOUSLY BE GOING FORWARD AND BACKWARD.

DATA ACKNOWLEDGED, it agreed.

USE THAT DATA TO EXPLAIN THE CONTRADICTIONS OF THE TWO POSITIONS YOU COMPUTED.

DATA INSUFFICIENT TO ARRIVE AT LOGICAL EXPLANATION, it answered.

ARE YOU CERTAIN THERE WAS NO ERROR IN YOUR CALCULATIONS?

THERE WAS NO ERROR.

DO YOU KNOW THAT IF WE DROPPED BACK INTO NORMAL SPACE, IT WOULD BE AT NEITHER OF THE POSITIONS YOU GAVE ME?

It replied with the characteristic single-mindedness: DATA SHOWS OUR TWO POSITIONS TO BE THOSE GIVEN.

He paused again. He was still getting nowhere while time fled by. How swiftly less than a hundred minutes could pass when they were all a man had left to him....

The computer was a genius with the mental initiative of a moronic child. It could find the answer for him but first he would have to take it by the hand and lead it in the right direction. To do that he would have to know more about the warp.

He wrote: EXPLAIN THE NATURE OF THE SPACE WARP AS SIMPLY AS POSSIBLE AND WITHOUT USING MATHEMATICS HIGHER THAN ALGEBRA.

It answered at once: THIS CANNOT BE DONE.

The chronometer read 12:30. He typed:

THIS SHIP WILL HAVE TO RETURN TO NORMAL SPACE NO LATER THAN 13:53. IT MUST BE MOVED TO A DIFFERENT POSITION WHILE STILL IN THE WARP.

DATA ACKNOWLEDGED, it replied.

THIS SHIP CANNOT OCCUPY TWO POSITIONS AT THE SAME TIME. YOUR MEMORY FILES SHOULD CONTAIN SUFFICIENT DATA TO ENABLE YOU TO FIND THE EXPLANATION OF THIS TWO-POSITIONS PARADOX. FIND THAT EXPLANATION.

SUBMIT METHOD OF PROCEDURE, it answered.

I DO NOT KNOW HOW. YOU WILL HAVE TO ARRIVE AT THE EXPLANATION UNAIDED.

THIS CANNOT BE DONE, it replied.

He wrote, with morbid curiosity:

IF YOU DO NOT FIND THE ANSWER UNAIDED YOU WILL BE DESTROYED ALONG WITH ME AT 13:53. DON'T YOU GIVE A DAMN?

It answered: GIVE A DAMN IS A SEMANTIC EXPRESSION I DO NOT UNDERSTAND. CLARIFY QUESTION.

He got out of the computer seat and walked about the room restlessly. He passed by the transdimensional viewscreen and communicator and pressed the communicator's signal button. A dial flickered in return, showing his signal was going out, but there was no sound in response. If only he could make contact with the brains in Observation—

He was umpty billion miles east of the sun and umpty billion miles west of the sun. He was racing faster than light in two different directions at once and he was sitting motionless under the blasters of two Slug cruisers.

Another thought came to him: even if he could move the ship while in the warp, where could he go?

He would have to go far beyond the outer limits of the solar system to escape detection by the Slug cruisers. And at that distance the sun would be only a yellow star, incapable of energizing the little solar power units. He would not live long after the last of the power was drained from the batteries and the air regeneration equipment ceased functioning. He would not even dare sleep, toward the last. There were no convection currents in the air of a ship without gravity, and it was imperative that the air be circulated constantly. The air circulation blowers would cease functioning while the ship still contained pure air but he would have to move about continually to breathe that air. Should he lie down to sleep he would smother to death in a carbon dioxide bubble of his own making.

If he managed to emerge into normal space at some point just outside Earth's atmosphere, beyond range of the cruisers, his driveless ship would descend as a blazing meteor. If, by some miracle, he could emerge into normal space just a few inches above the space-field it would be to materialize into space already occupied by air. Such a materialization would be simultaneously fatal to him and to the electronic components of the shuttle and computer.

And if he did not move the ship, the Slug cruisers would disintegrate him. He had four hypothetical choices of his way to die, all equally unpleasant.

He smiled wanly at his reflection in the bright metal bordering the viewscreen and said, "Brother—you've had it!"

He went to the control room, there to brush his fingers across the useless control buttons and look into the viewscreen that revealed only black and limitless Nothing.

What was the warp? Surely it must have definite physical laws of some kind. It was difficult to imagine any kind of existence—even the black nothing of the warp—as being utterly without rule or reason. If he knew the laws of the warp he might find some means of survival hitherto hidden from him.

There was only one way he could learn about the warp. He would have to question the computer and continue questioning it until he learned or until his time was up.

He returned to the computer and considered his next question. The computer had calculated their positions from observations of the sun and other stars in front of the ship—what would similar calculations based on observations of the stars behind the ship reveal? He typed:

USE FIRST THE TRIANGULATION METHOD AND THEN THE SPECTRUM-SHIFT METHOD TO DETERMINE OUR POSITION FROM OBSERVATIONS MADE OF THE STARS OF OPHIUCHUS.

The answers appeared. They showed the ship to be simultaneously speeding away from Ophiuchus and toward it.

He asked: DO THESE TWO POSITIONS COINCIDE WITH THOSE RESULTING FROM THE OBSERVATIONS OF ORION?

YES, it answered.

Was the paradox limited to the line of flight?

He asked the computer: WHAT IS OUR POSITION, COURSE AND SPEED AS INDICATED BY THE STARS AT RIGHT-ANGLES TO OUR FORWARD-BACKWARD COURSE; BY THE STARS OF URSA MINOR AND CRUX?

The answer appeared on the panel: the ship was racing sideward through the warp in two diametrically opposed directions, but at only one-third the speed with which it was racing forward and backward.

So now the ship had four impossible positions and two different speeds.

He frowned at the computer, trying to find some clue in the new data. He noticed, absently, that the hand of one of the dials was near zero in the red section of the dial. He had not noticed any of the dials registering in the danger zone before....

He jerked out of his preoccupation with apprehension and typed: TELL ME IN NON-TECHNICAL LANGUAGE THE MEANING OF THE HAND NEAR ZERO ON THE DIAL LABELED MAX. *ET. REF.*

It answered: ONE OF MY CIRCUITS WAS DAMAGED BY THE SUDDEN RELEASE OF AIR PRESSURE. I WILL CEASE FUNCTIONING AT THE END OF FOUR MORE MINUTES OF OPERATION.

He slammed the master switch to OFF. The lights on the board went out, the various needles swung to zero, leaving the computer a mindless structure more than ever resembling an overgrown refrigerator.

Four minutes more of operation ... and he had so many questions to ask before he could hope to learn enough about the warp to know what he should do. He had wasted almost an hour of the computer's limited life, leaving it turned on when he was not using it. If only it had told him ... but it was not the nature of a machine to voluntarily give information. Besides, the receding hand of the dial was there for him to see. The computer neither knew nor cared that no one had thought it worthwhile to teach him the rudiments of its operation and maintenance.

It was 12:52. One hour and one minute left.

He put the thought aside and concentrated on the problem of finding the key to the paradox.

What conceivable set of circumstances would cause receding stars to have a spectrum shift that showed them to be approaching the ship? Or, to rephrase the question, what conceivable set of circumstances would cause approaching stars to appear to dwindle in size?

The answer came with startling suddenness and clarity:

There was no paradox—the ship was expanding.

He considered the solution, examining it for flaws of logic, and found none. If he and the ship were expanding the wave length of light would diminish in proportion to the increasing size of the retinas of his eyes and the scanner plates of the transdimensional viewscreens: would become shorter and go into the ultraviolet. At the same time, the increasing size of himself and the ship would make the Earth and sun relatively smaller and therefore apparently receding.

The same theory explained the two different speeds of the ship: its length was three times its diameter so its longitudinal expansion would proceed at three times the speed of its cross-sectional expansion.

Everything checked.

How large was the ship now?

He made a rough calculation and stared almost unbelievably at the results. He was a giant, more than a third of a light-year tall, in a ship that was six light-years long and two light-years in diameter. Far Centauri, which had required thirty years to reach in the fastest interplanetary ship, floated seventy-one feet away in the blackness outside the hull.

And the sun and Earth were in the room with him, going into the shuttle's silvery focal ball.

He would have to ask the computer to make certain his theory was valid. His time was too critically short for him to waste any of it with speculation based on an erroneous theory.

He switched on the computer and it lighted up again. He typed rapidly:

ASSUME THIS SHIP TO BE MOTIONLESS AND EXPANDING
WOULD THAT THEORY SATISFACTORILY EXPLAIN ALL THE
HITHERTO CONTRADICTORY PHENOMENA?

There was a brief pause as the computer evaluated its data, then it answered with one word:

YES.

He switched it off again, to squander none of its short period of usefulness until he had decided upon what his further questions should be. At last, he had some grounds for conjecture; had learned something about the warp the designers of the shuttle had not suspected. Their calculations had been correct when they showed a ship would travel in the warp at many times the normal space speed of light. But somewhere some little factor had been overlooked—or never found—and their precise mathematics had not indicated that the travel would be produced by expansion.

Nature abhors a vacuum. And the black, empty warp was a vacuum more perfect than any that existed in normal space. In the normal space universe there were millions of stars in the galaxy and millions of galaxies. In the warp there was utter Nothing. Did the physical laws of the warp demand that matter be scattered throughout it, in emulation of its rich neighbor in the adjoining dimension? Was the warp hungry for matter?

He rejected the thought as fantasy. There was some explanation that the physicists would eventually find. Perhaps there was a vast size-ratio difference between the two dimensions; perhaps the warp was far larger than the normal space universe and some co-universal law demanded that objects entering it become proportionally larger.

None of that aspect of his circumstances, however, was of importance. There was only one prime problem facing him: how to move the ship within less than an hour to some point in the warp where his emergence into normal space would result in neither instant nor days-away death and where he would have the time to try to carry out the responsibility, so suddenly placed in his hands, of delivering the space warp shuttle to the *Thunderbolt*.

The long-range task depended upon his immediate survival. He had to move the ship, and how did a man move a driveless ship? It might not require a very large propulsive force—perhaps even an oxygen tank would serve as a jet. Except that he had none.

He could use part of the air in the ship. Its sudden release should move the ship. There was a sun very near: Alpha Centauri. If he had the proper tools,

and the time, he could cut a hole in the hull opposite Centauri ... but he had neither the tools nor the time.

And what good would it do him if he could emerge into normal space at the desired distance from Centauri? He would be provided with power for the air regenerators by the solar power units but not power sufficient to operate the shuttle. He would breathe, and eat, for a week. Then the small amount of food on the ship would be gone and he would breathe for another four or five weeks. And then he would die of starvation and his driveless ship would continue its slow drift into the sun, taking his bones and the shuttle with it.

He would have to go to Sirius and he would have to reach it the first try or never. If he could emerge into normal space at the proper distance from Sirius he would have power from it to operate the communicator. The *Thunderbolt* would come at once when it received his message and swallow the little *Argosy* in its enormous hold. The return to Earth would be the swift one through the warp and the Slug cruisers, so bold in pursuit of unarmed interplanetary ships, would quickly cease to exist.

At 13:53 Sirius would be somewhere in or near the bow of the ship. The ship would not have to be moved more than two thirds of its length—twenty meters. He could do that by releasing part of the air in the shuttle room through the sternward airlock.

How much air?

He tried to remember long-forgotten formulas. So many cubic feet of air at such and such a pressure when released through an opening of such and such a diameter would exert a propulsive force of.... Hell, he didn't know. And not even the computer would be able to tell him because there were so many unknown factors, such as the proportion of the ship's mass lost to the Slug blasters, the irregular shape of the airlock opening, the degree of smoothness of its metal....

He made calculations with pencil and paper. He would have to move the ship with extreme precision. A light-hour short of the proper distance put him too far from the sun for it to power the communicator, a light-hour beyond put him in the sun's flaming white heart. One light-hour out of eight point six light-years was approximately one part out of seventy-five

thousand. He would have to move the ship with an accuracy of point eight three centimeters—one hundredth of an inch.

One hundredth of an inch!

He laid the pencil back down, almost numbly. He could never open and close an airlock and move a mass of thousands of tons with an accuracy of a hundredth of an inch. The very thought was wildly fantastic.

He was already far closer to Sirius than he would be if he tried to get any closer. And that was over eight light-years from it.

He looked at the chronometer and saw the hands had already reached 13:20. Thirty-three minutes left to him. Sirius was near—soon it would be in the bow of the ship—and Sirius was eight point six light-years away.

How could he move the ship a certain distance accurate to one hundredth of an inch? He couldn't. The answer was blunt and ugly and irrefutable: he couldn't.

He got up and walked across the room, feeling like a man who had in quick succession been condemned, reprieved, recondemned. He had been projected into a situation for which he had had no preliminary training whatever; had been made sole custodian and operator of a computer and a space warp shuttle that he had never before been permitted to touch. He had used the sound but not at all brilliant mind nature had given him to solve the riddle of the paradoxes and learn where he was and where he wanted to go. He had done quite well—he had solved every problem of his survival and the shuttle's delivery except the last one!

He passed by the shuttle and stopped to rest his hand on the bright, silvery focal ball. The solar system would be deep inside the ball; the atoms of the ball larger than Earth, perhaps, and far more impalpable than the thinnest air. The Slug cruisers would be in there, infinitesimally tiny, waiting for him to return....

No—faulty reasoning. The solar system was as it had always been, not diminished in size and not really in the ball. It was only that two different points in two different dimensions coincided in the ball....

He saw the answer.

He did not have to move the ship to Sirius—he had only to move the ball!

There would be little time, very little time. First, to see if the warp shuttle was portable—

It was. When he unfastened the clamp that held it to the stand it lifted up freely, trailing a heavy cable behind it. He saw it was only a power supply cable, with a plug that would fit one of the sockets in the bow of the ship. He left the shuttle floating in the air, leashed by the cable, and went to the computer. Next, he would have to know if Sirius would be fully in the ship —

He switched the computer on and typed:

DETERMINE THE DISTANCE FROM THE CENTER OF THE WARP SHUTTLE'S FOCAL BALL TO THE SPACE WARP POSITION OF SIRIUS AT 13:53, BASING YOUR COMPUTATIONS ON THE EXPANDING-SHIP THEORY.

It gave him the answer a moment later: 18.3496 METERS.

He visualized the distance, from his knowledge of the ship's interior, and saw the position would be within the forward spare-parts room.

Next, to learn exactly where in that room he should place the shuttle. He could not do so by measuring from the present position of the shuttle. The most precise steel tape would have to be at exactly the right temperature for such a measurement to be neither too short nor too long. He had no such tape, and the distance from the focal ball was only part of the necessary measuring: he would have to measure off a certain distance and a precisely certain angle from the purely imaginary central line of the ship's axis to intersect the original line. Such a measurement would be impossible in the time he had.

He considered what would be his last question to the computer. The hand was touching the zero and his question would have to be worded very clearly and subject to no misinterpretations. There would be no follow-up questions permitted.

He began typing:

IT IS DESIRED THAT THIS SHIP EMERGE INTO NORMAL SPACE ONE LIGHT-HOUR THIS SIDE OF SIRIUS AT 13:53. THIS WILL BE ACCOMPLISHED BY MOVING THE WARP SHUTTLE TO SUCH A POSITION THAT ITS FOCAL CENTER WILL BE IN A SPACE WARP POSITION COINCIDING WITH A NORMAL SPACE POSITION ONE LIGHT-HOUR THIS SIDE OF SIRIUS AT 13:53. CONSIDER ALL FACTORS THAT MIGHT HAVE AFFECTED THE DIMENSIONS OF THIS SHIP, SUCH AS TEMPERATURE CHANGES PRODUCED BY OUR NORMAL SPACE ACCELERATION AND DECELERATION, WHEN COMPUTING THE POSITION OF SIRIUS. THEN DEFINE THAT LOCATION IN RELATION TO THE STRUCTURAL FEATURES OF THE ROOM'S INTERIOR. DO THIS IN SUCH A MANNER THAT PLACING THE SHUTTLE IN THE PROPER POSITION WILL REQUIRE THE LEAST POSSIBLE AMOUNT OF MEASURING DISTANCES AND ANGLES.

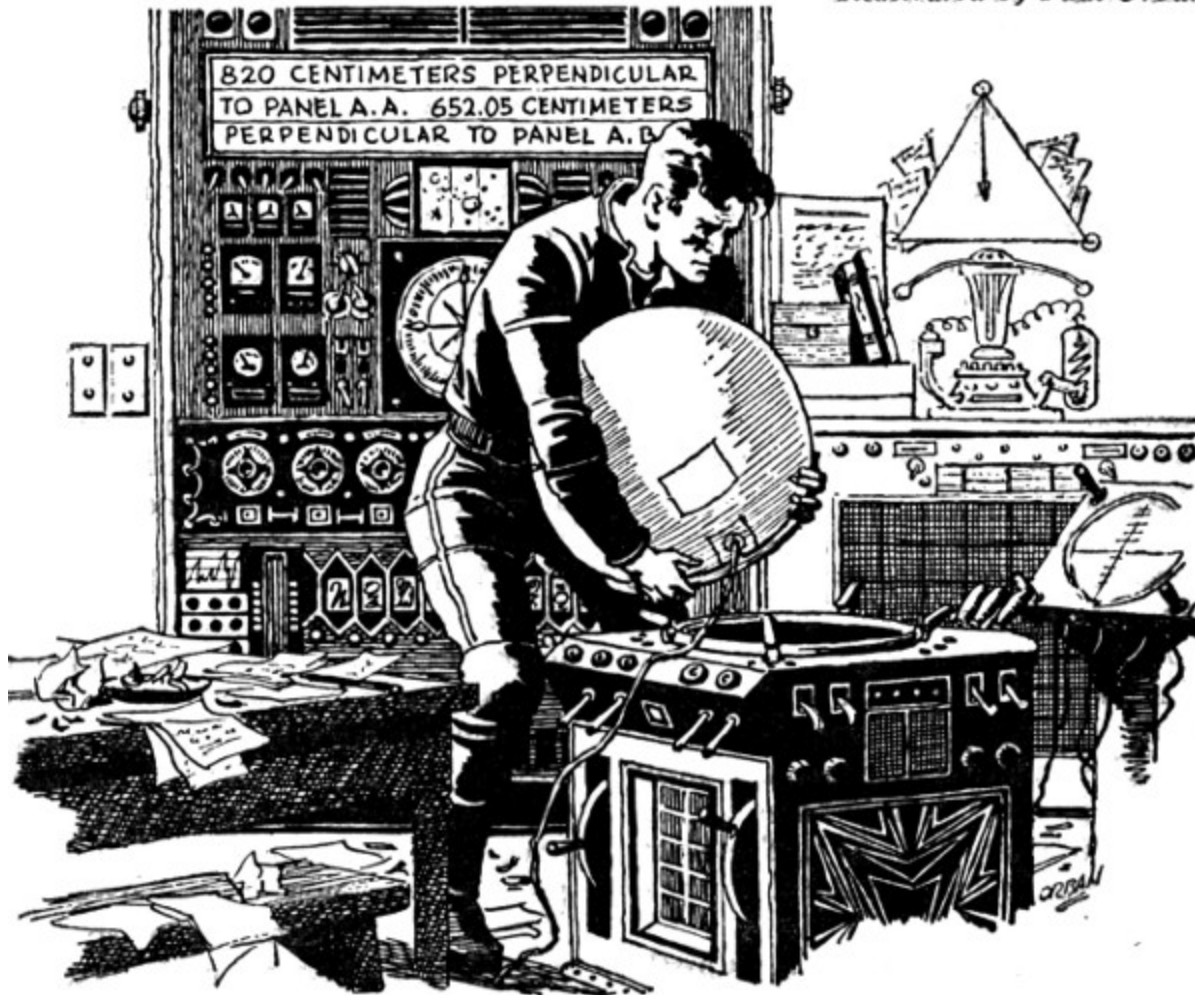
It seemed to take it an unduly long time to answer the question and he waited restlessly, unpleasantly aware of the hand touching zero and wondering if the computer's mind was baffled by the question; the mind that thought best in terms of orderly mathematics and could not know or care that measurement by protractor and tape would result in a position fatally far from that described by the neat, rigid figures.

Then the answer appeared, beautifully concise:

POSITION WILL BE IN CORNER OF ROOM, 764.2 CENTIMETERS ABOVE FLOOR PLATE, 820 CENTIMETERS PERPENDICULAR TO PANEL AA, 652.05 CENTIMETERS PERPENDICULAR TO PANEL AB.

The computer died with an oddly human sigh. Its last act had been to give him the location of Sirius in such a manner that he could accurately position the shuttle's focal ball with the aid of the precision measuring devices in the ship's repair room.

He went to the shuttle and picked it up in his arms. It was entirely weightless, and each magnet-clicking step he took toward the bow of the ship brought Sirius almost half a light-year nearer.



He squinted against the white glare of Sirius in the viewscreen as he continued his terse report to the *Thunderbolt's* commander: "I have about a week's supply of food. How long will it be until you reach me?"

The commander's reply came after the pause caused by the distance involved:

"We'll be there within three days. Go ahead and eat hearty. But how did you travel from Earth to Sirius in only two hours? My God, man—what kind of a drive did that ship have?"

"Why, it didn't have any drive from the start," he said. "To get here I"—he frowned thoughtfully—"you might say I walked and carried the ship."

*** END OF THE PROJECT GUTENBERG EBOOK BRAIN TEASER

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